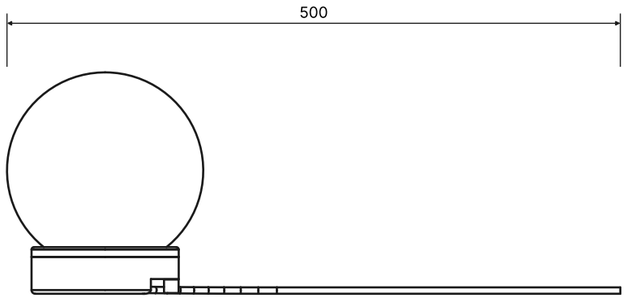
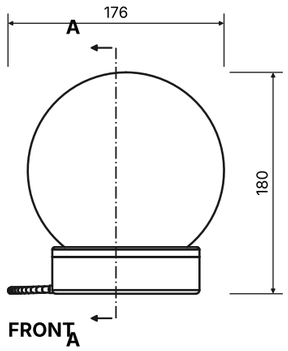
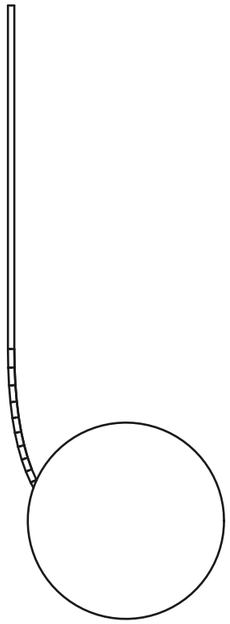
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

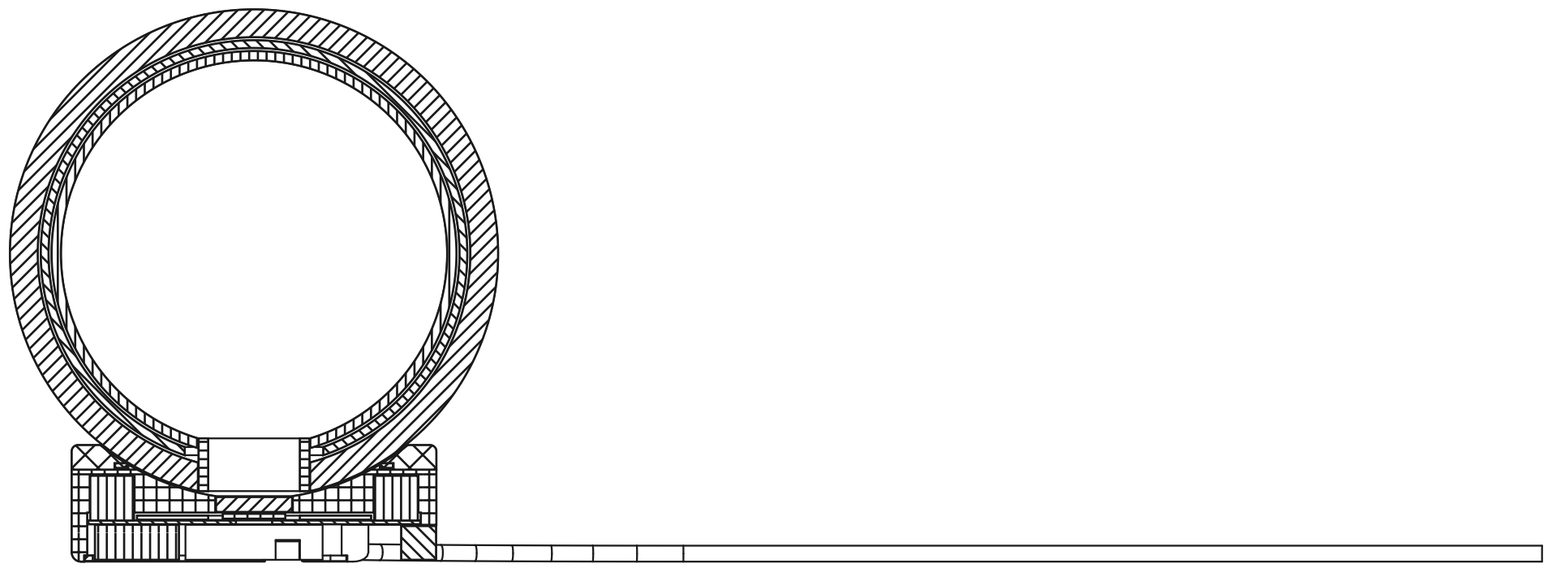
General arrangement

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Section A-A (sheet 1): the product cut at x -8.1 and seen from the right; cut faces hatched, a different angle for each part.



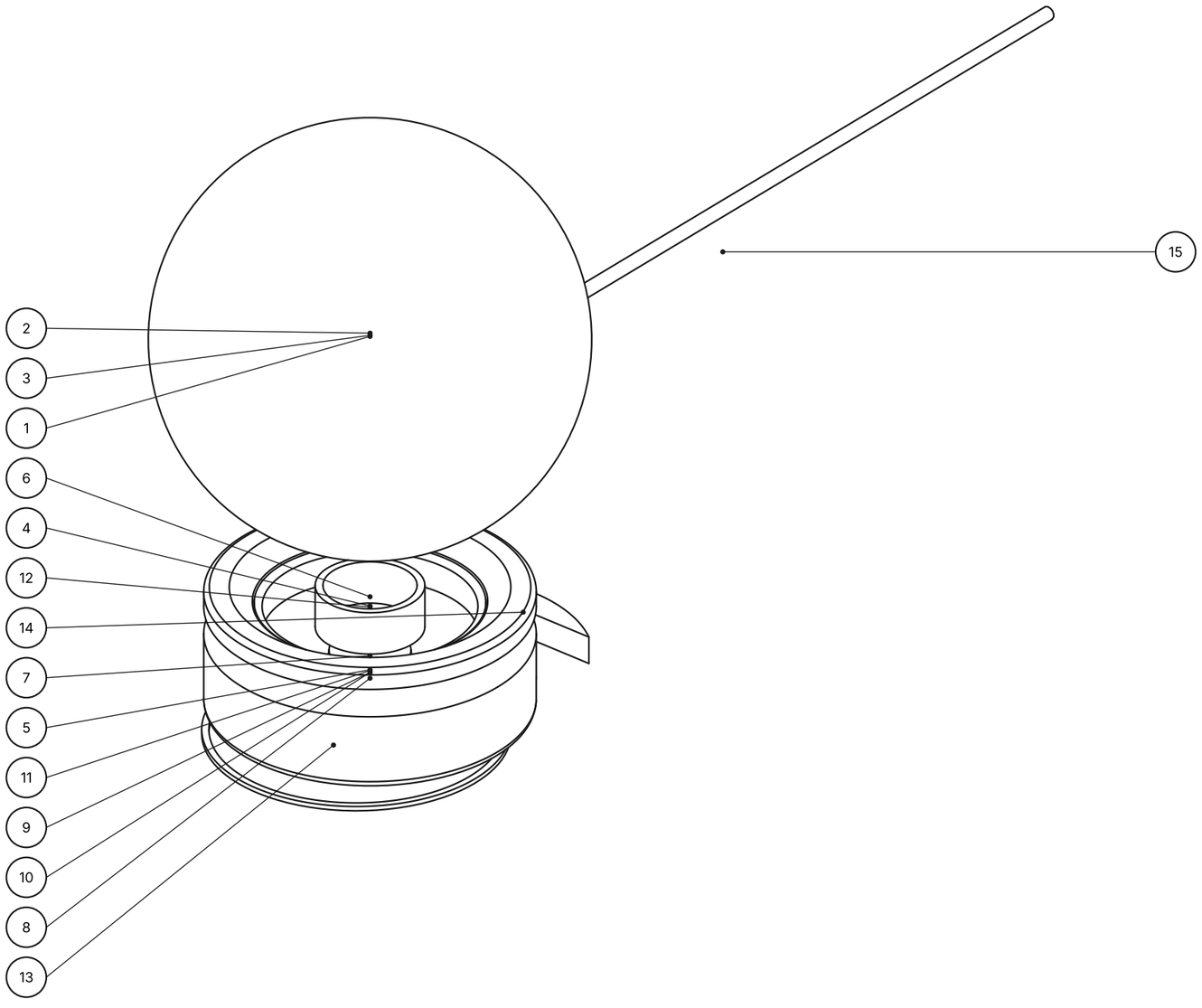
Section A-A

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

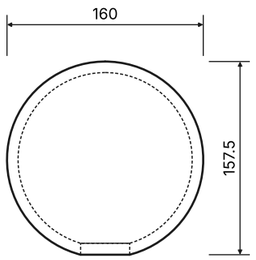
mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

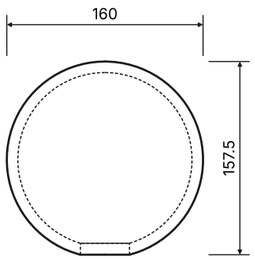
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: lathe, cast-acrylic; 634 cm3 each; 1 for the set; 1:5

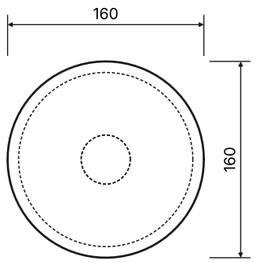
the ball's outside: two hemispheres turned from cast block in the tint, 9 wall, bonded at the equator and polished (or rotationally cast: ask the caster); a 40 port at its underside



FRONT



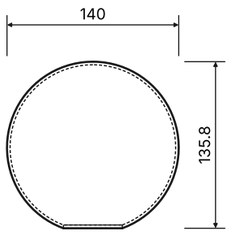
FROM THE RIGHT



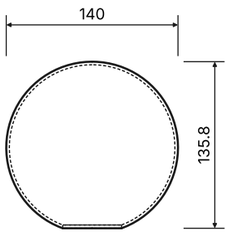
FROM ABOVE

2. liner: lathe, cast-acrylic; 144 cm3 each; 1 for the set; 1:5

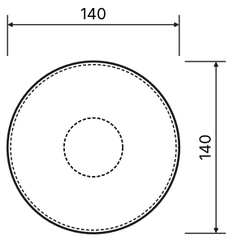
the diffuser: a 2.5 ball in clear, bead-blasted, in two halves inside the shell



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

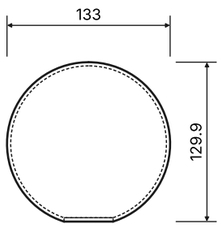
mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

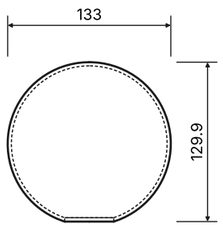
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. skin: lathe, cast-acrylic; 155 cm3 each; 1 for the set; 1:5

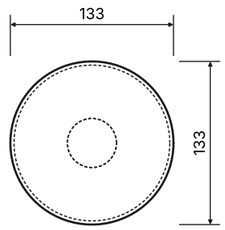
a 3 ball in the tint inside the liner: seen through the port, the frost is still behind acrylic



FRONT



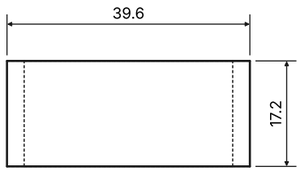
FROM THE RIGHT



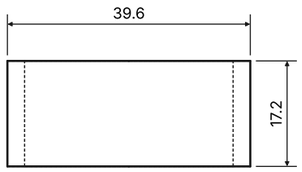
FROM ABOVE

4. grommet: lathe, aluminium; 6 cm3 each; 1 for the set; 1:1

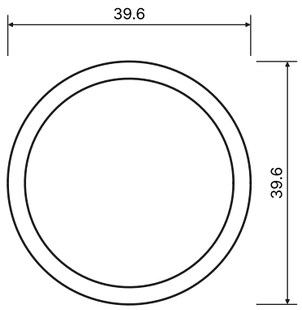
black anodised aluminium: lines the port and closes the three layers there, bonded



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

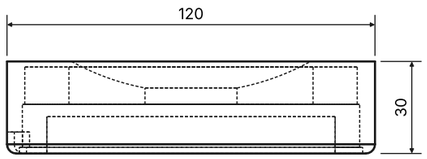
mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

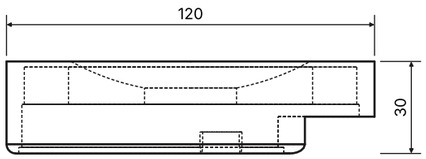
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. base: lathe, cast-acrylic; 100 cm3 each; 1 for the set; 1:2

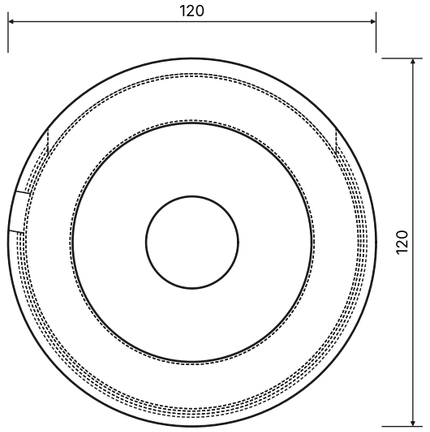
black cast acrylic, polished: 120 × 30, the cup turned in it, the pocket (110 × 16) from below, the bay and the cable notch milled



FRONT



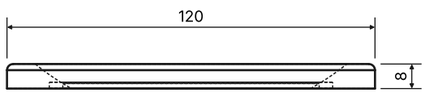
FROM THE RIGHT



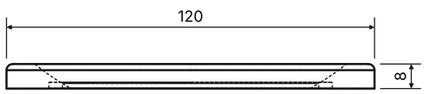
FROM ABOVE

6. ring: lathe, aluminium; 36 cm3 each; 1 for the set; 1:2

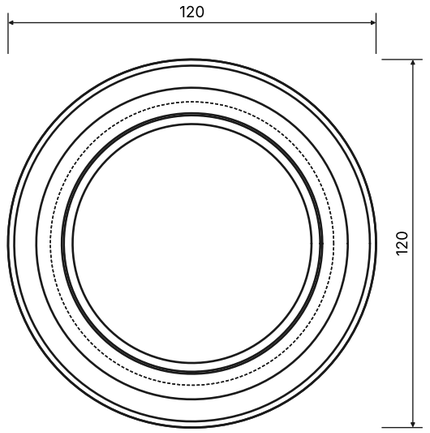
aluminium, black anodised, satin, insulated from the chassis: the touch (tap to switch, hold to dim, 2200 to 2700 K); metal, as it must be to sense a touch



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

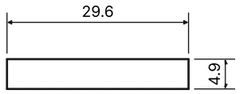
mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

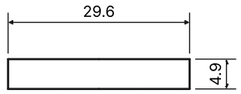
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

7. window: lathe, cast-acrylic; 3 cm3 each; 1 for the set; 1:1

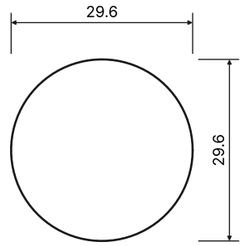
a clear acrylic plug, sealed with an O-ring: the LED under it, the base IP44 with the ball off



FRONT



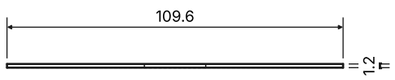
FROM THE RIGHT



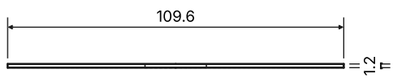
FROM ABOVE

8. chassis: mill, aluminium; 11 cm3 each; 1 for the set; 1:2

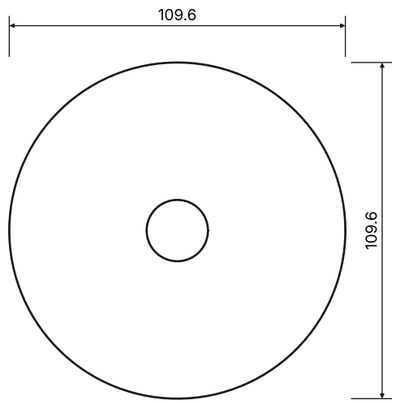
aluminium 6061 plate: carries the board, the LED's heat path and the USB-C receptacle; screwed up into the acrylic on threaded inserts, an O-ring between



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

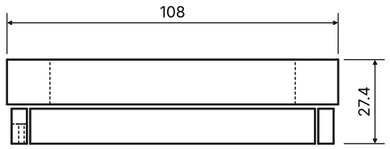
mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

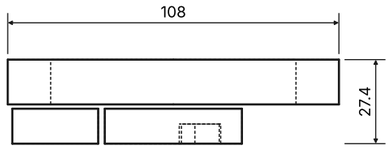
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

9. ballast: mill, steel; 85 cm3 each; 1 for the set; 1:2

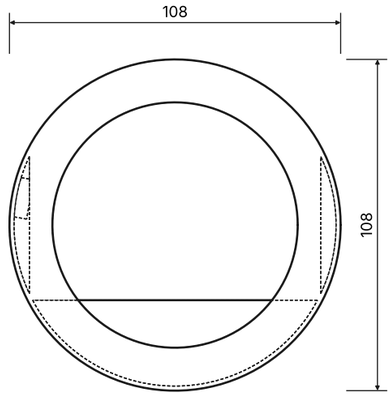
mild steel, zinc-plated: the weight under the ball (the steel drum's job before): a plate round the bay and a ring above the chassis, under the cup; screwed through the chassis



FRONT



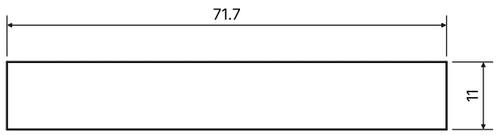
FROM THE RIGHT



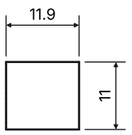
FROM ABOVE

14. blank: mill, cast-acrylic; 6 cm3 each; 1 for the set; 1:1

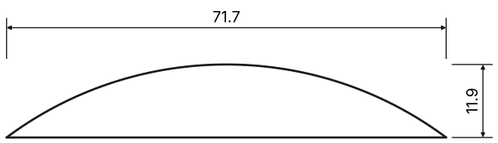
the bay's blanking cap, black acrylic polished, flush with the drum: two magnets hold it



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	lathe	1	Cast acrylic tube, tinted through	170 × 170 × 167	358.58
2	liner	lathe	1	Cast acrylic tube, tinted through	150 × 150 × 146	256.18
3	skin	lathe	1	Cast acrylic tube, tinted through	143 × 143 × 140	239.45
4	grommet	lathe	1	Aluminium 6061	50 × 50 × 27	46.67
5	base	lathe	1	Cast acrylic tube, tinted through	130 × 130 × 40	112.04
6	ring	lathe	1	Aluminium 6061	130 × 130 × 18	52.61
7	window	lathe	1	Cast acrylic tube, tinted through	40 × 40 × 15	45.93
8	chassis	mill	1	Aluminium 6061	120 × 120 × 11	49.01
9	ballast	mill	1	Mild steel	118 × 118 × 37	50.21
10	board	buy	1	to make (a contract PCB house) The Orb's board: a USB-C PD sink (9 V, 20 W class; STUSB4500 kind), the LED driver (two channels, 2200 and 2700 K warm-dim, about 2.5 W), the capacitive touch dimmer on the anodised ring, and the cartridge's port (5		24.00
11	led	buy	1	Nichia or Seoul (to choose) A 30 mm round warm-dim LED board, two channels (2200 K and 2700 K, CRI 90+), about 2.5 W at full, 200 lm; a second 4-LED ring at the column's foot to light the column (0.3 W)		9.00
12	seat	buy	1	A silicone seat ring 92 × 84 × 1.5 (Shore 40, black) in the base's cup: the ball rests on it, and it stops a nudge rolling it		1.50
13	foot-ring	buy	1	Two EPDM O-rings (the window, the chassis to the drum), a silicone foot ring 110 x 2 (Shore 40, non-slip), a sealed USB-C receptacle (IP67 rated, e.g. GCT USB4135 kind): IP44		5.00
14	blank	mill	1	Cast acrylic tube, tinted through	82 × 22 × 21	46.50
15	cable	buy	1	any (made to order) A slim black braided-fabric USB-C to USB-C cable, 2 m, 3 A, a black right-angle overmoulded plug for the notch		6.00
16	adapter	buy	1	any (UL/CE listed) A 20 W USB-C PD wall adapter, black, UL 62368-1 / CE listed (bought listed, so the plug-in Orb is a low-voltage lamp)		7.00
17	cartridge	buy	1	[option: cartridge] to make (a battery-pack house) The battery cartridge: one Li-ion pouch cell about 8 × 60 × 90 (6,000 mAh at 3.7 V, about 22 Wh), its protection board and fuel gauge, in a black polished acrylic and PC housing 93 × 83 × 11 with		38.00
18	cartridge-pads	buy	1	[option: cartridge] Mill-Max (or alike) Two gold-plated flat pads under the cartridge, for the staff tray's spring pins		2.00

Parts list

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. A crystal ball: a 160 tinted ball, low in the cup of a 120 × 38 base. The ball lifts off: the LED is under a sealed window in the base, so balls swap colour by hand, as the Dome's covers do.
2. Plug-in as standard: USB-C PD in through a notch in the underside at the back left, so the base sits flat; a slim black fabric cable (2 m) and a 20 W adapter in the box; no battery. Touch the black ring: tap to switch, hold to dim, 2200 to 2700 K as it dims.
3. Cordless as an option: a battery cartridge (one 8 × 60 × 90 Li-ion pouch, about 22 Wh, with its own protection board) slides into the bay under the base from the back and latches on two magnets and a detent, flush with the drum. About 10 hours at the bar setting; it charges in the lamp over the USB-C, or on the staff tray through two gold pads on its underside. Without it a blanking cap closes the bay.
4. IP44: the base sealed by the window's and the chassis's O-rings, the cartridge's own contacts sealed; with the ball on or off. Wipe with a damp cloth.
5. Black, so the tint is the only colour: the drum black cast acrylic, polished, over an aluminium chassis and a steel ballast; the touch ring and the port's lining black anodised aluminium.

Notes

Orb table lamp: Plug-in table lamp in cast acrylic for bars, a crystal ball, cordless with its battery ca

mm | issue 2026-10-20 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15